

Experiment - 03

- Aim: Create the cube with suitable dimension and fact tables based on ROLAP, MOLAP and HOLAP model.

- Theory:

- (1) Introduction -

A data cube is a multidimensional data structure used in OLAP (Online Analytical Processing systems) to support analytical queries in Business Intelligence environments. It is built using fact tables and dimension tables organized in a star or snowflake schema.

- (2). The three main types of OLAP - ROLAP, MOLAP, and HOLAP differ in how they store and process data for analysis.

- (a) Relational Online Analytical Processing (ROLAP)

ROLAP servers are placed between relational backend server and client front end tools. It uses relational or extended DBMS to store and manage warehouse data. ROLAP has basically 3 main components: Database server, ROLAP server, front end tool.

Advantages of ROLAP -

- Rolap is used to handle large amount of data.
- ROLAP tools don't use pre calculated data cubes.
- Data can be stored efficiently.
- ROLAP can leverage functionalities inherent in the relational database.

◦ Disadvantages of ROLAP -

- Performance of ROLAP can be slow.
- In ROLAP, it is difficult to maintain aggregate tables.
- Limited by SQL functionalities.

(b). Multidimensional Online Analytical Processing (MOLAP)

- MOLAP does not use relational database to storage. It stores in optimized multidimensional array storage. The storage utilization may be low with multidimensional data stores. Many MOLAP server handle dense and sparse datasets by using two levels of data storage representation levels. MOLAP has 3 components: Database server, MOLAP server and front end tool.

◦ Advantages of MOLAP -

- Used for complex calculations.
- MOLAP is optimal for operations such as slice and dice.
- MOLAP allows fastest indexing to the pre-computed summarized data.

- Disadvantages of MOLAP -

- MOLAP can't handle large amount of data.
- In MOLAP, requires additional investment.
- Without re aggregation, difficult to change dimension.

(c). Hybrid Online Analytical Processing - (HOLAP)

- Hybrid is a combination of both ROLAP and MOLAP. It offers functionalities of both ROLAP and as well as MOLAP like faster communication computation of MOLAP and higher scalability of ROLAP. The aggregations are stored separately in MOLAP store. Its server allows storing the large data volumes of detailed information.

- Advantages of HOLAP -

- HOLAP provides the functionalities of both MOLAP and ROLAP.
- HOLAP provides fast access at all levels of aggregation.

- Disadvantages of HOLAP -

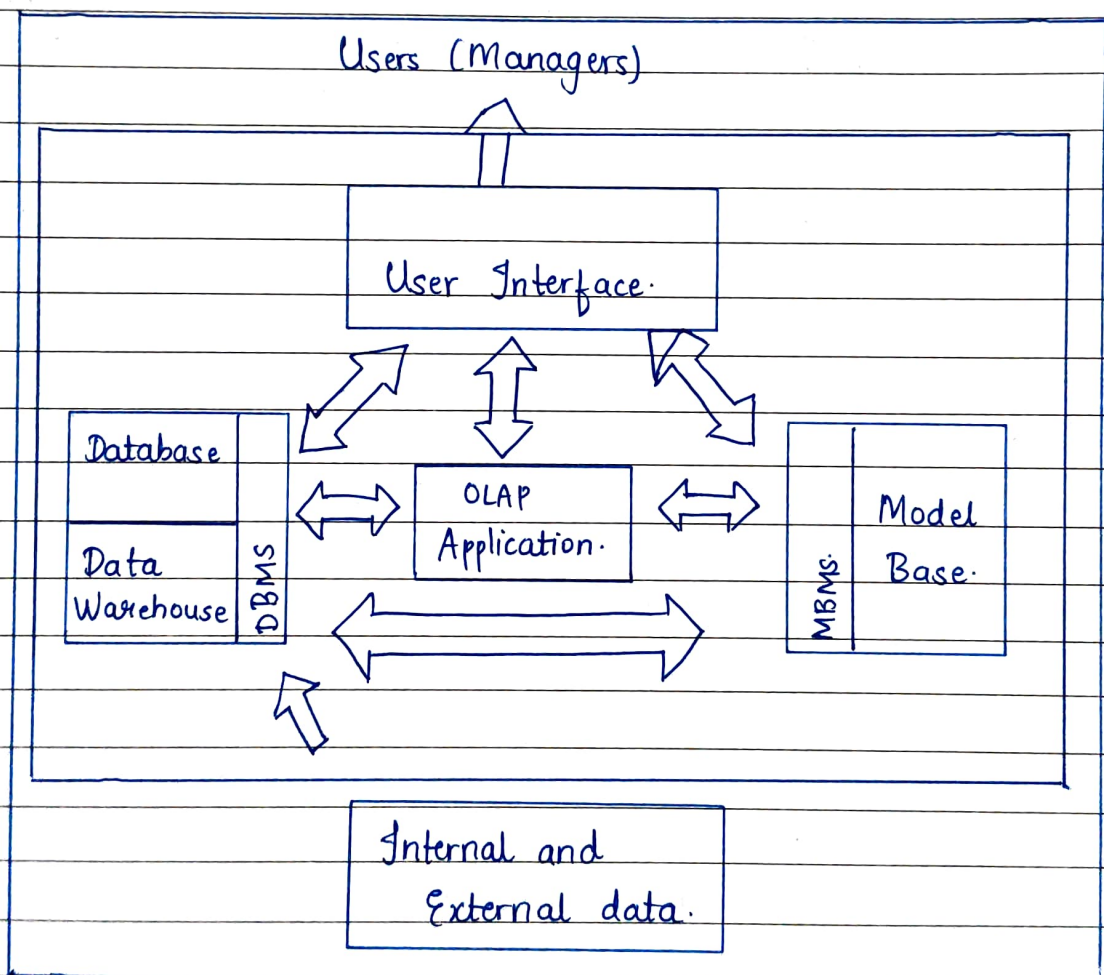
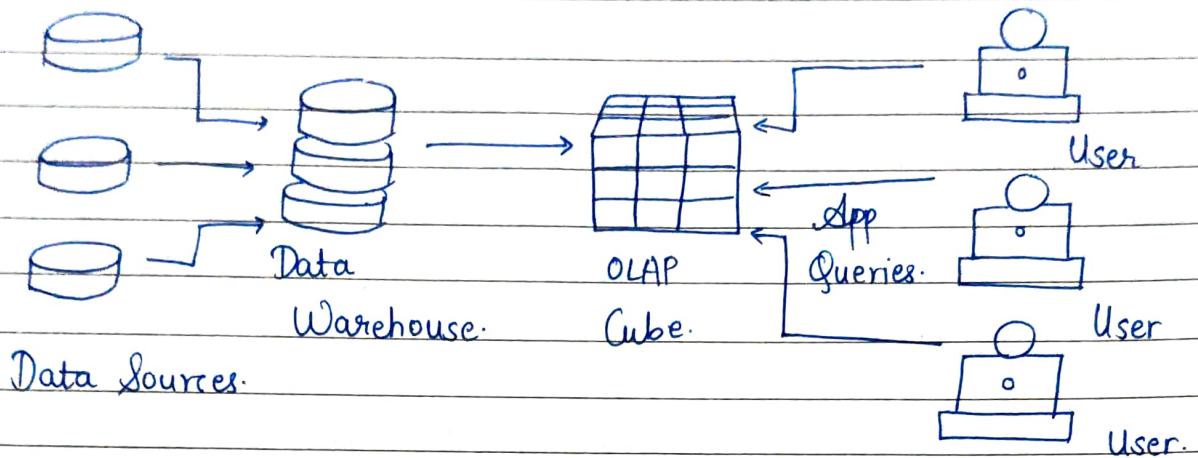
- HOLAP architecture is very complex to understand because it supports both MOLAP and ROLAP.
- More complex to manage.
- Data sync between sources can be challenging.

◦ Difference between ROLAP, MOLAP and HOLAP.

Basis.	ROLAP	MOLAP	HOLAP.
◦ Storage location for Aggregation.	Relational Database is used for storage location of summary aggregation	Multidimensional database is used as storage location	Multidimensional database is used for storage location.
◦ Processing time.	Processing time of ROLAP is very slow.	Processing time of MOLAP is fast.	Processing time of HOLAP is fast.
◦ Storage space requirement	Large storage space requirement in ROLAP as compare to MOLAP, HOLAP.	Medium storage space in MOLAP compared to other two.	Small storage space requirement in HOLAP as compare to MOLAP and ROLAP.
◦ Latency.	Low latency in ROLAP as compare to MOLAP and HOLAP	High latency in MOLAP compared to ROLAP, HOLAP.	Medium latency in HOLAP.
◦ Query response time.	Slow query response time in ROLAP as compare to MOLAP, HOLAP.	Fast query response time in MOLAP.	Medium query response time in HOLAP.
◦ Storage location for detail data.	Relational database for storage of detail data	Multidimensional database for detail data storage.	Relational database is used as storage location for detail data.

THE OLAP PROCESS.

How data is prepared for OLAP.



The model selection depends on -

- Data Volume.
- Query performance requirements.
- Storage availability.
- Business Intelligence objectives.

CONCLUSION:

Thus, we studied and implemented the OLAP models of ROLAP, MOLAP and HOLAP for business intelligence application.

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